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Earth's Sky: The Sun and Earth

Day, Night, and Seasons

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Introduction

The Sun and the Earth are the two objects every human being has known, in the most literal sense, forever. One is a star, a hydrogen sphere a million times the volume of our planet, fusing its substance into light at a rate that has held steady for four billion years. The other is a cooled cinder of that same primordial material, at exactly the right distance to keep water liquid on its surface, wrapped in a magnetic field that turns the star's violence into auroras instead of annihilation. This book is about the relationship between them, which turns out to be a single physical system: the Sun-Earth system, the coupled machinery of light, wind, fields, and orbit that makes one planet habitable and keeps the entire question of life in play.

The relationship was the first subject of astronomy and remains one of the busiest. The Sun gave humanity its calendar, its clock, its direction-finder, and its gods; the Earth's motion around it was the argument that created modern science; and in the last century the pair acquired a third dimension, the space between them, which turned out to be not empty at all but filled with the star's own exhaled atmosphere, a wind that blows past our planet continuously and a magnetic canopy that shields us from it. The study of that in-between is now its own discipline, with its own fleet of spacecraft, and the practical stakes are modern to the core: power grids, satellites, aviation, and every radio link on Earth depend on the weather that the Sun sends.

The chapters that follow keep the two bodies together, which is the honest way to treat them. A book only about the Sun would drift toward the stellar physicists' problems, fusion and magnetism and the corona's heat, and a book only about the Earth would drift toward the geologists' ones, but the interesting questions live in the coupling: how the star's light sets the planet's climate, how its wind shakes the planet's field, how its rhythm of spots structured the climate record, and how the planet's own record, in ice and tree rings, became a scientific archive of the star's behavior. The Sun and the Earth are also the calibration pair for the whole search for life elsewhere: the "habitable zone" that exoplanet hunters speak of is defined by asking, planet by planet, whether it has what our planet has, at the distance our planet has it, from a star like our star.

One caution belongs up front, and it is the book's most practical lesson. The Sun-Earth relationship is a physical system with a stability we tend to take for granted, because the star's output, the planet's orbit, and the magnetic shielding have all held steady for the span of civilization. But the same system has modes, storms, and long-term swings, and the modern world has built its infrastructure into the assumption that the modes never engage. They do. The star's own record, and the planet's, are the evidence, and learning to read them is where the story has been going for two centuries and where it arrives next.

Chapter 1: The Discovery of The Sun and Earth

The discovery of the Sun-Earth system took two and a half millennia, and its stages are the great landmarks of science: first the geometry, then the physics, then the connection.

The geometric stage began with the Greek refusal to leave the sky sacred. Anaxagoras, in the fifth century BC, proposed that the Sun was a fiery stone, larger than the Peloponnese, and that the Moon shone by reflected light, a claim so corrosive to civic religion that he was prosecuted for impiety; the intellectual pattern of the next two thousand years was set in that trial. Aristarchus of Samos, a century later, used the Moon's quarter phase to estimate the Sun's distance as some eighteen to twenty times the Moon's, badly low but structurally correct: the Sun was vastly farther, and therefore vastly larger, than the Earth, an argument he drew to its Copernican conclusion a millennium and a half early. Eratosthenes, the same era, measured the Earth itself, its circumference from the shadow angles at two Egyptian cities on the same noon, to within a few percent by most modern reckonings. The sizes were known, in rough, astonishingly early; the arrangement, and the physics, were not.

The geometry stage also fixed the relationship's annual texture, the seasons, which the Greeks analyzed correctly, the tilt hypothesis being attested by the second century BC and its obliquity measured by Eratosthenes' successors to a fraction of a degree. The same program produced the analemma's ancestor, the figure-of-eight track of the Sun's noontime position in the sky that a year of watching records, the construction Ptolemy systematized and that every sundial's correction curve descends from; the relationship's small asymmetries, the equation of time, the tilt's coupling with the orbit's slight eccentricity, were catalog work of the ancient world, and the medieval and Islamic astronomers' tables refined them, so the byproduct of the discovery stage was an exact civil description of the light channel, the where-and-when of the Sun in the sky, known centuries before the what of the light itself.

The arrangement came with the heliocentric argument, made by Aristarchus, made again by Copernicus in 1543, and clinched by Galileo's telescope in 1610 and the phases of Venus. The discovery stage that matters most to this book's subject, though, is subtler than the fight over the center: it is the realization that the Sun is a physical object and the Earth is a physical object, and both can be measured. The distance to the Sun, the astronomical unit, the yardstick of the solar system, was the prize. Aristarchus's method gave poor leverage; the transits of Venus, predicted by Kepler's tables and pursued in the famous eighteenth-century expeditions, gave a planet-crossing geometry that fixed the unit to within one percent; and the Mars parallax campaign of 1672, Cassini and Richer measuring from Paris and Cayenne simultaneously, had produced a respectable value a century earlier. By the end of the nineteenth century the astronomical unit was known to a fraction of a percent, and the scale of the Sun-Earth relationship, 149.6 million kilometers, give or take, was a working number.

One further geometric discovery belongs to this stage, because it completed the pair's connection in a literal sense: the speed of light. In 1676, Ole Rømer, working from Picard's observations of Jupiter's moons at the Paris Observatory, noticed that the eclipses of Io ran late when the Earth was far from Jupiter and early when it was near, and drew the correct conclusion, that light travels at a finite speed, crossing the diameter of the Earth's orbit in about 22 minutes by his figures. The speed itself was refined over the following centuries, but the discovery's import for this book's subject was immediate: the Sun-Earth relationship is a signal path, eight minutes wide, and every astronomical observation is a look into the past by exactly the width of the path.

The physical stage followed the same arc as all of nineteenth-century astronomy: spectroscopy. When the Sun's light was dispersed and read, the signature of the star's substance appeared, hydrogen above all, with helium the standing puzzle until its Earth-bound isolation in 1895; and when thermodynamics matured, the question of the Sun's energy supply became the century's great physical problem. The geologists, Darwin's son among them, arguing from the sediment record for an Earth hundreds of millions of years old, collided with the physicists, Kelvin above all, arguing that a Sun burning any known fuel could not keep burning for more than a few dozen million years. The collision was productive: it meant the Sun's energy was unknown physics, and the unknown held until the twentieth century supplied first the nuclear atom, then Eddington's argument that stellar cores fuse hydrogen, then Bethe's 1938 reaction chain. The Sun's light was finally explained as nuclear, and the Sun-Earth relationship acquired its deepest fact: every calorie the Earth receives is fusion's exhaust, eight minutes and twenty seconds in transit.

A number attaches to the geometric stage's finish and belongs in the record: the astronomical unit, once the transits' prize, is now defined exactly, 149,597,870,700 meters, by international agreement, because radar ranging of the planets made the old triangulations obsolete and the unit became a definition rather than a measurement. The definition's precision, meters where the transit expeditions achieved thousands of kilometers, is the geometric discovery stage's closing achievement, and it is worth remembering what the precision is for: every orbit in the solar system, every spacecraft trajectory, and every parallax that measures the stars begins from this one number, the length of the thread between the book's two subjects.

The connection stage, the twentieth century's contribution, is the discovery that the space between is not empty. The Sun exhales a wind, a permanent outflow of plasma, predicted by Eugene Parker in 1958 from the corona's implied behavior and confirmed by Mariner 2's instruments at Venus distance in 1962; the wind carries the Sun's magnetic field outward as a spiraling canopy, and the Earth's own field carves a cavity in it. The connection stage is also where this book's applied story lives, because the wind is not steady: it carries gusts, and the gusts, when they reach Earth, are the space weather that disturbs the planet's magnetic environment. The discovery timeline closes with the recognition that the pair is a system, with inputs, outputs, and a record: the star's behavior preserved in the planet's ice and trees, and the planet's response preserved in the starlight-driven chemistry of both. The chapters that follow work through the system piece by piece.